

Implication of $f(Q, T)$ gravity on the formation of the charge compact stars and constraining their mass-radius relations in lower mass gap region

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ABSTRACT

In this paper, we explore a charged isotropic stellar model within the framework of $f(Q, T)$ gravity. By deriving solutions to the modified field equations, we systematically analyze the physical characteristics of the stellar configuration to assess its relevance and stability. Our results show that the electric charge distribution adheres to key criteria of regularity and finiteness. Notably, the charge behavior is significantly affected by the charge parameter σ and the coupling parameter m , while demonstrating stability in response to variations in n . We find that the surface charge ranges from 1.85×10^{20} C to 3.31×10^{20} C, which is consistent with prior research, highlighting the crucial role of electric charge in maintaining stellar stability against gravitational collapse. Additionally, our analysis of density and isotropic pressure reveals both quantities to be finite and regular at the core, effectively eliminating the presence of singularities. The central density aligns well with typical neutron star configurations, while the isotropic pressure approaches zero at the boundary. Investigating the causality conditions, we confirm that the sound speed remains subluminal, and the adiabatic index $\Gamma(r)$ exceeds the critical threshold of $\frac{4}{3}$, indicating stability throughout the stellar configuration. The equilibrium of forces, evaluated through modified TOV equations, illustrates a harmonious balance among gravitational, electrostatic, and hydrostatic forces. The relationship between mass and radius indicates a maximum mass ranging from $1.83 M_{\odot}$ to $2.99 M_{\odot}$, with radii that are consistent with observational data. In conclusion, our findings underscore the substantial impact of the charge parameter σ in enhancing the compactness and stability of higher-mass compact stars, providing support for the existence of stars within the mass gap region.

1. Introduction

Astrophysical structures essentially form the systematically organized galaxies that weave the cosmic web. To understand the fundamental properties of the galaxy, one needs to understand the intricate physics of these astrophysical objects. Several cosmological studies suggest that stars, composed of self-gravitating luminous plasma, originated from the expansion of superheated matter-energy. Post big bang, mass clumps mainly made of hydrogen and helium contracted under im-

mense gravity, triggering nuclear fusion. Because of that, the strong inward gravitational pull was counteracted by the outward pressure to maintaining equilibrium until the hydrogen fuel depleted. The resulting imbalance caused gravitational collapse, giving rise to ultra compact objects such as black holes, neutron stars, white dwarfs and so on. Chandrasekhar (1936) was the first to emphasize on this process in the context of relativistic astrophysics, establishing that the compact stars are stabilized by maintaining balance between inward gravity and outward nuclear pressure. Meanwhile, collapsing dust spheres were studied

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by Oppenheimer and Snyder (1939) predicting the formation of black holes. In another work, the role of anisotropic pressure in collapse was examined by Misner and Sharp linking mass and density with heat flux. Shear-free collapsing fluids were explored by Glass (1981) and Santos (1985) and junction conditions were derived, which was used by Herrera and Santos (2004) to explore dissipative collapse indicating energy loss via radiation and heat.

One of the ways to model stars is to consider them as self-gravitating fluid spheres. An exact solution was proposed by Krori and Barua (1975) using this idea. Moreover, considering the isotropic matter distribution and general relativity (GR) framework, the compactness of static spheres composed of ordinary fluid is constrained by the Buchdahl limit (Buchdahl, 1959). However, this constraint changes when there is the presence of electric charge (Andreasson, 2008). Some works, such as Maurya and Tello-Ortiz (2019) considered the existence of electric charge and obtained accurate analytical solutions to the Einstein-Maxwell field equations that describe charged anisotropic compact stars. Strong gravitational attractive can be counterbalanced by repulsive Coulomb forces as well, indicating that the stellar evolution is significantly impacted by electromagnetic fields. The existence of charge can thus enhance stability in strong gravity. The effect of charge on dynamical instability was explored by Glazer (1976), while it was shown by Zhang et al. (1982) that when effect of charge matches those of mass density, the structural properties of neutron star structure alters. Slowing of collapse due to the presence of charge was shown by Bekenstein (1971) which was later backed by Esculpi and Aloma (2010) for anisotropic spheres. The charge can also hinder the possibility of Reissner-Nordström black hole formation as shown in Pinheiro and Chan (2013). The collapse in charged systems happens slower than the uncharged ones, as shown by Sharif and collaborators (Sharif and Abbas, 2011; Sharif and Azam, 2012; Sharif and Yousaf, 2013; Sharif and Bhatti, 2015). Promising approaches encompass various cosmological and quantum implications, providing diverse frameworks for further exploration in future studies (Razina et al., 2011; Esmakhanova et al., 2012; Errehymy et al., 2024b; Koussour et al., 2024a,b; Akramov et al., 2024a,b; Koussour et al., 2025).

The mass-range of the compact objects have been debated since several decades. Chandrasekhar first derived the limiting mass of white dwarfs. He established the well-known Chandrasekhar limit of approximately $1.4M_{\odot}$ (Chandrasekhar, 1931). Several theoretical studies backed by observational evidence further supported this value (Shapiro and Teukolsky, 1983). This estimation revolves around the idea of the degenerate pressure of neutron stars. However, because of recent precise measurements and gravitational wave detection, this paradigm has shifted, as we now have a broader range of neutron star masses. Binary neutron star merger GW170817 analysis provided the upper limit to be $M_{\max} \leq 2.17M_{\odot}$ (Margalit and Metzger, 2017). Further studies indicated that the equation of state of neutron matter must be stiff enough to allow maximum masses exceeding $2M_{\odot}$, opening doors for the possibility of long-lived post-merger residues for binary systems with total masses above $2.73M_{\odot}$ (Shibata et al., 2017). The absence of a relativistic optical counterpart in GW170817 implies a likely maximum neutron star mass of approximately $2.15 - 2.25M_{\odot}$ (Ruiz et al., 2018; Rezzolla et al., 2018). The gravitational wave event GW190425 further suggested a binary neutron star system with a total mass of around $3.4M_{\odot}$ (Abbott et al., 2020a). Another event, GW190408, indicated the presence of a neutron star with a mass between $2.5 - 2.65M_{\odot}$, merging with a $26M_{\odot}$ black hole (Abbott et al., 2020b). These findings suggest that the traditional assumption of a standard $1.4M_{\odot}$ neutron star mass is increasingly questionable. Several reasons can be there for the presence of higher mass neutron stars, such as the presence of exotic components, considering rotation and so on. Often modified gravity theories support the existence of higher mass neutron stars in a much more efficient way than their GR counterpart. For example, GW190814 event's secondary component was analyzed as a quark star (QS) with a color-flavor-locked (CFL) equation of state (EOS) under $f(R, T)$ gravity by Tangphati et al. (2022) and they found that the upper mass and radius limits are affected

by the curvature coupling enabling QS masses reaching $2.86M_{\odot}$ and a radius of 12.43 km. Anisotropic quark stars were studied in Tangphati et al. (2021) under Einstein-Gauss-Bonnet (EGB) gravity, yielding masses up to $2.83M_{\odot}$.

As discussed earlier, extended and modified gravity theories often produce results that are both intriguing and physically significant, including the prediction of compactness, and mass-radius relation. As a result, they are worth exploring in the analytical modelling of compact stars. Additionally, the theoretical foundations of these alternative approaches are highly fascinating. The development of GR by Einstein (1915); Hilbert (1915); Einstein (1916) influenced not only physics and cosmology but also mathematics. Utilizing Riemannian geometry, GR describes spacetime through a metric tensor and an affine connection, with the Einstein tensor capturing geometric and gravitational properties. Over the last few decades, the Λ CDM model has provided a consistent framework for explaining numerous cosmological observations within the geometry of GR. Since the universe is expanding faster than ever before and the cosmological constant Λ has its limits, scientists are looking into other models to explain dark energy and dark matter. In this context, it is worthwhile to mention that shortly after GR's inception, Weyl (1918) proposed extending Riemannian geometry to unify gravity and electromagnetism, where spacetime and non-metricity were used as sources of electromagnetic fields. Initially, this was dismissed by Einstein, but this laid the groundwork for future developments. Another big step forward in differential geometry was the discovery of torsion (Cartan, 1922). This led to the Einstein-Cartan theory (Cartan, 1923, 1924, 1925), which connects torsion to the spin density of matter (Hehl et al., 1976). A unique concept termed as teleparallelism was introduced by Weitzenböck (1923), where the space is characterized by zero curvature and non-zero torsion. Later on, Einstein applied these structures to propose a teleparallel theory unifying gravity and electromagnetism (Einstein, 1928). The teleparallel equivalent of GR (TEGR) (Möller, 1961; Pellegrini and Plebanski, 1963; Hayashi and Shirafuji, 1979) evolved into $f(T)$ gravity, where the curvature is replaced by torsion scalar in field equations. This formulation simplifies field equations to second-order differentials, distinguishing it from fourth-order equations in $f(R)$ gravity (Harko and Lobo, 2018). Teleparallel theories have been widely used in cosmology and astrophysics to explain cosmic acceleration without invoking dark energy (Ferraro and Fiorini, 2007, 2008; Bengochea and Ferraro, 2009; Linder, 2010; Boehmer et al., 2012; Harko et al., 2014; Farrugia et al., 2018; D'Agostino and Luongo, 2018; Fontanini et al., 2019).

In another approach, non-metricity Q is used to describe gravity, which is known as symmetric teleparallel gravity. This theory was proposed by Nester and Yo (1999), and it was extended by Beltrán Jiménez et al. (2018) to obtain $f(Q)$ gravity because of its reliance on the dark components. In Maurya et al. (2022), the compact stars of masses comparable to GW190814 event were identified in $f(Q)$ gravity. In a more generalized version of $f(Q)$ gravity, the possible polarization of GWs in Minkowski spacetime and the velocity propagation were studied in Hohmann et al. (2019). In the last few years, a significant amount of works have explored in $f(Q)$ theories (Hohmann, 2021; Ambrosio et al., 2022; Jiménez et al., 2020; Atayde and Frusciante, 2021; Jiménez et al., 2019; D'Ambrosio et al., 2022; Zhao, 2022a; De and How, 2022; Lin and Zhai, 2021; Frusciante, 2021; Khyllap et al., 2021). A new matter-geometry coupling framework, known as $f(Q, T)$ gravity, was recently introduced by Xu et al. (2019b). In this formulation, an arbitrary function of the non-metricity scalar Q and the trace of the energy-momentum tensor, T are used to express the Lagrangian. Although significant efforts have been made, but the majority of studies (aside from the original proposal) have centered on observational analysis rather than deepening the theoretical understanding of the framework, which is still in its nascent phase. Additionally, a detailed review indicates the absence of a fully developed covariant formulation, along with missing terms in the energy balance equation. A specific class of $f(Q, T)$ gravity was introduced by considering that the scalar non-metricity follows the Weyl

form and is depicted by a vector field (Xu et al., 2020). In this case, the field equations were derived using a variational principle with the condition that the total scalar curvature vanishes, enforced via a Lagrange multiplier. Various cosmological models within this framework have been studied, with predictions cross-checked against the Λ CDM model and observational studies. Various cosmological and astrophysical implications of $f(Q, T)$ gravity have been explored in the following work (Nájera and Fajardo, 2022). The authors (Kausar et al., 2025; Maurya et al., 2025b) also delve into the fascinating realm of strange quark stars by exploring a type of space-time that is both spherically symmetric and static, along with an anisotropic fluid. They employ Heintzmann metric components to uncover solutions to the field equations of their modified $f(Q, T)$ theory of gravity, working with two different models that incorporate both linear and nonlinear terms. Interestingly, this isn't the only research focused on strange quark stars in modified gravity theories. Other studies have looked into similar concepts using frameworks like $f(R, T)$ gravity (Banerjee et al., 2025b; Goswami et al., 2025; Errehymy et al., 2024a; Maurya et al., 2025a; Errehymy et al., 2022; Singh et al., 2020; Maurya et al., 2019; Malik et al., 2024b), $f(R, \phi, X)$ (Malik et al., 2025; Malik, 2024; Malik et al., 2024a,c) gravity, Rastall gravity (Tangphati et al., 2024a; Banerjee et al., 2025a; Mustafa et al., 2022), rainbow gravity (Tangphati et al., 2025b, 2024b; Errehymy et al., 2025), energy-momentum squared gravity (Banerjee et al., 2025c), and $f(R, L_m, T)$ gravity (Tangphati et al., 2025a; Mota et al., 2024; Pretel, 2024). These works often explore quark matter equations of state, which are influenced by the bag constant, and consider both interacting and non-interacting scenarios, all while taking into account the latest astrophysical observations.

While the challenges of working with pseudo-Riemannian spacetimes have been addressed within the modified gravity theories, several contemporary issues in cosmology remain, particularly with the STEGR. For instance, integrating the STEGR with the Λ -CDM model still leads to the cosmological constant problem. Thus, exploring extensions within the framework of symmetric teleparallel gravity is beneficial. Similar to $f(R)$ gravity, we can extend this theory by generalizing $Q \rightarrow f(Q)$. These theories, proposed in Beltrán Jiménez et al. (2018), yield generalized Friedmann equations that may account for the Universe's accelerated expansion through geometric terms rather than a cosmological constant. Additionally, we can explore further extensions to these $f(Q)$ theories. By incorporating the trace of the stress-energy tensor with the non-metricity scalar into the Lagrangian, we arrive at $f(Q, T)$ gravity (Buchdahl, 1959). Theories that couple geometry with matter are particularly intriguing, as they disrupt stress-energy conservation (Xu et al., 2019b; Harko et al., 2011). This disruption can lead to remarkable outcomes, including the transfer of stress-energy between matter and geometry, as well as particle production and annihilation (Xu et al., 2019b; Wu et al., 2018), which may be related to quantum gravity. A key feature of $f(Q, T)$ gravity is its coupling of geometric quantities with matter properties, enabling exploration of how matter influences spacetime curvature. Similar traits are found in $f(R)$ and $f(T)$ theories, where modifications to the Einstein-Hilbert action introduce terms that affect the universe's dynamics. This coupling can yield new insights into gravity's behavior. These modified theories tackle significant cosmological issues like dark energy, cosmic acceleration, and structure formation. By directly linking geometry and matter, $f(Q, T)$ gravity provides pathways to address problems in standard cosmological models, such as the cosmological constant problem, framing the universe's expansion in geometric terms. Mathematically, these theories often use similar techniques, including generalized Friedmann equations for cosmic evolution. In certain cases, equations from $f(Q, T)$ gravity can reduce to those of $f(R)$ or $f(T)$, highlighting their interconnected nature. This relationship emphasizes their shared goal of enhancing our understanding of gravitational behavior. Additionally, the phenomenological predictions of $f(Q, T)$ gravity may offer insights into gravitational interactions similar to those in $f(R)$ and $f(T)$ models. Potential deviations from standard cosmological behavior could be tested against observational

data, providing empirical avenues to validate these modified theories (Nájera and Fajardo, 2021, 2022). Overall, the connections between $f(Q, T)$ gravity and other modified gravity theories enrich gravitational research and deepen our understanding of the universe.

In the realm of $f(Q, T)$ gravity, charged compact stars also offer an intriguing window into the mysteries of our universe. These stellar remnants, which include neutron stars and other exotic objects, can showcase properties that differ from what we expect based on GR. For instance, the relationship between a star's mass and radius can shift due to the modifications introduced by this theory. Observations from instruments like NICER (Neutron star Interior Composition Explorer) allow scientists to measure neutron star radii with remarkable precision. By comparing these measurements to theoretical predictions, researchers can gain insights into the nature of gravity itself, potentially uncovering new physics that challenges our current understanding, gravitational waves provide another exciting avenue for exploration. When compact stars merge, they create ripples in spacetime that can be detected by observatories like LIGO and VIRGO. Analyzing these gravitational waves helps researchers infer details about the stars' internal structures and the effects of modified gravity. Charged compact stars might also emit unique electromagnetic signals due to their electric fields, which can be picked up by telescopes like the Five-hundred-meter Aperture Spherical Telescope (FAST) or X-ray observatories like Chandra. These emissions can reveal vital information about a star's properties, such as its charge and magnetic characteristics.

The study of quasi-normal modes (QNMs) represents another exciting aspect of this research. These oscillations can provide clues about the stars' behavior in modified gravity, and their detection through gravitational waves could offer a direct way to test theories like $f(Q, T)$. Additionally, thermal emissions from the surfaces of these stars can help scientists understand their cooling rates, which may be influenced by the modifications in this framework. Looking ahead, the future of this research is bright. Upcoming observatories, such as the James Webb Space Telescope (JWST) and the Square Kilometre Array (SKA), will greatly enhance our ability to observe these fascinating objects across various wavelengths. By combining data from gravitational waves with electromagnetic observations, researchers will be able to build a more comprehensive picture of compact stars and the fundamental nature of gravity. This exciting journey into the cosmos not only deepens our understanding of the universe but also challenges us to rethink what we know about the very fabric of reality.

In $f(Q, T)$ gravity, the shift from neutron stars to dirty black holes—black holes that are surrounded by matter—can be understood through the complex interactions between modified gravity effects and external matter fields (Matyjasek, 2020; Bamber et al., 2021; Torres et al., 2023; Soroushfar et al., 2024; Noori Gashti et al., 2025; Turimov et al., 2025; Rakhmanov et al., 2025). This transition is especially important for models that suggest a maximum neutron star mass of about 2.99 solar masses. When neutron stars reach these high masses, they are on the brink of collapsing into black holes. In this scenario, the presence of nearby matter, like accretion disks along with the changes in spacetime geometry due to $f(Q, T)$ gravity, can significantly impact the collapse dynamics and the characteristics of the resulting compact objects. The external matter fields around dirty black holes, such as accretion disks or dark matter halos, interact with the modified gravitational field in $f(Q, T)$ gravity. This interaction occurs through the relationship between non-metricity Q and the trace of the energy-momentum tensor T . This relationship can alter the effective curvature of spacetime and the distribution of stress-energy around these compact objects, potentially speeding up or slowing down the collapse. For neutron stars nearing their maximum mass, gravitational interactions with surrounding matter may also affect their stability, determining whether they remain intact or transition into dirty black holes. Furthermore, these external fields could leave observable marks on gravitational waves produced during these transitions, such as shifts in phase or changes in amplitude due to altered tidal effects. In $f(Q, T)$ gravity, dirty black holes might

show unique characteristics compared to those described by GR. For example, external matter could change their effective mass and spin, or lead to noticeable differences in phenomena like weak deflection angles or the behavior of accretion disks. These distinct features could provide a way to differentiate neutron stars from dirty black holes in modified gravity theories, particularly for objects in the mass-gap region. Observations of gravitational waves, along with other signals like X-rays from accretion disks, could help explore these transitions and validate the predictions of $f(Q, T)$ gravity in comparison to standard GR.

In this article, within the $f(Q, T)$ gravity framework, we obtain exact solutions for charged compact stars and conduct a thorough physical analysis, including stability considerations and the mass-radius relationship. We also look to investigate the possible upper limit of compact star mass in $f(Q, T)$ gravity. Additionally, we determine the radii of certain compact objects with known masses. The article is organized as follows: Section 2 provides a brief review of $f(Q, T)$ gravity, whereas section 3 discusses modified field equations in $f(Q, T)$ gravity. The matching conditions for the astrophysical system have been discussed in Section 4, while the physical analysis has been done in Section 5, and lastly, Section 6 provides concluding remarks.

2. A brief review of $f(Q, T)$ gravity

In $f(Q, T)$ gravity, the generalized action integral is expressed as Xu et al. (2019a),

$$S = \int \sqrt{-g} \left[\frac{1}{2} f(Q, T) + \mathcal{L}_M + \mathcal{L}_e \right] d^4x, \quad (1)$$

with matter and electromagnetic field Lagrangian densities being denoted by $\mathcal{L}_M$ and $\mathcal{L}_e$ and are associated with the energy-momentum tensors $T_{\mu\eta}$ and $E_{\mu\eta}$. Here, the metric tensor determinant is denoted by g and the gravitational sector is expressed as Θ . For simplicity, we adopt the convention $8\pi G = c = 1$ throughout this work.

The non-metricity tensor corresponding to the affine connection is expressed as,

$$Q_{\kappa\mu\eta} \equiv \nabla_\kappa g_{\mu\eta} = \partial_\kappa g_{\mu\eta} - \Gamma^\delta_{\kappa\mu} g_{\delta\eta} - \Gamma^\delta_{\kappa\eta} g_{\mu\delta}, \quad (2)$$

where, the standard affine connection is denoted by $\Gamma^\delta_{\kappa\mu}$ and is formulated as,

$$\Gamma^\delta_{\kappa\mu} = \left\{ \begin{matrix} \delta \\ \kappa\mu \end{matrix} \right\} + K^\delta_{\kappa\mu} + L^\delta_{\kappa\mu}, \quad (3)$$

where the deformation tensor, Levi-Civita connection and contorsion tensor are denoted as, $K^\delta_{\kappa\mu}$, $\left\{ \begin{matrix} \delta \\ \kappa\mu \end{matrix} \right\}$, $L^\delta_{\kappa\mu}$ respectively. They are mathematically represented as,

$$\left\{ \begin{matrix} \delta \\ \kappa\mu \end{matrix} \right\} = \frac{1}{2} g^{\delta\varphi} (\partial_\kappa g_{\varphi\mu} + \partial_\mu g_{\delta\eta} - \partial_\delta g_{\kappa\mu}), \quad (4)$$

$$K^\delta_{\kappa\mu} = \frac{1}{2} T^\delta_{\kappa\mu} + T_{(\kappa}{}^\delta{}_{\mu)}, \quad (5)$$

$$L^\delta_{\kappa\mu} = \frac{1}{2} Q^\delta_{\kappa\mu} - Q_{(\kappa}{}^\delta{}_{\mu)}, \quad (6)$$

with $T^\delta_{\kappa\mu}$ being the torsion tensor. The superpotential $P^\varphi_{\mu\eta}$, another key element in the STEGR formalism is expressed as,

$$P^\varphi_{\mu\eta} = -\frac{1}{2} L^\varphi_{\mu\eta} + \frac{1}{4} (Q^\varphi - \tilde{Q}^\varphi) g_{\mu\eta} - \frac{1}{4} \delta^\varphi_{(\mu} Q_{\eta)}, \quad (7)$$

where, the following represents the non-metricity tensor trace,

$$Q_\varphi = Q^\mu_{\varphi\mu}, \quad \tilde{Q}_\varphi = Q^\mu_{\varphi\mu},$$

eventually, the non-metricity scalar is introduced as,

$$Q = -Q_{\varphi\mu\eta} P^{\varphi\mu\eta}. \quad (8)$$

The $f(Q, T)$ gravity field equations are retrieved by using variational principle through varying the action (1) with respect to the inverse metric tensor $g^{\mu\eta}$ as,

$$\begin{aligned} \frac{2}{\sqrt{-g}} \nabla_\varphi \left(f_Q \sqrt{-g} P^\varphi_{\mu\eta} \right) + f_Q \left(P_{\mu\varphi\kappa} Q^\varphi_{\eta}{}^{\kappa} - 2 Q^\varphi_{\mu}{}^{\kappa} P_{\varphi\kappa\eta} \right) \\ + \frac{1}{2} f g_{\mu\eta} = -(T_{\mu\eta} + E_{\mu\eta}) + f_T (T_{\mu\eta} + \Phi_{\mu\eta}), \end{aligned} \quad (9)$$

with $f_Q = \frac{\partial f(Q, T)}{\partial Q}$, $f_T = \frac{\partial f(Q, T)}{\partial T}$, and $T_{\mu\eta}^{eff} = T_{\mu\eta} + \alpha \Theta_{\mu\eta}$. The conventional energy-momentum tensor $T_{\mu\eta}$ along with the extra source $\Theta_{\mu\eta}$ is expressed as,

$$T_{\mu\eta} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} \mathcal{L}_M)}{\delta g^{\mu\eta}}, \quad (10)$$

$$E_{\mu\eta} = \frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} \mathcal{L}_e)}{\delta g^{\mu\eta}}. \quad (11)$$

The hyper-momentum tensor, an additional unknown quantity in Eq. (9) is stated as $\Phi_{\mu\eta}$ and is expressed as $\Phi_{\mu\eta} = g^{\varphi\beta} \frac{\delta T_{\varphi\beta}}{\delta g^{\mu\eta}}$.

Moreover, the additional constraint can be retrieved by using Eq. (1) and can be described as,

$$\nabla_\mu \nabla_\eta (\sqrt{-g} f_Q P^\varphi_{\mu\eta}) = 0. \quad (12)$$

The curvature-free and torsion-free conditions (Beltrán Jiménez et al., 2018) are satisfied by the affine connection,

$$\Gamma^\varphi_{\mu\eta} = \left(\frac{\partial x^\varphi}{\partial \xi^\alpha} \right) \partial_\mu \partial_\eta \xi^\alpha. \quad (13)$$

Choosing a specific coordinate system, known as the co-incident gauge, ensures that the affine connection satisfies $\Gamma^\varphi_{\mu\eta} = 0$. Consequently, the non-metricity equation simplifies to,

$$Q_{\kappa\mu\eta} \equiv \nabla_\kappa g_{\mu\eta} = \partial_\kappa g_{\mu\eta}. \quad (14)$$

This approach streamlines computations as now the primary dynamical variable is nothing but the metric. However, the action loses diffeomorphism invariance except within the STGR framework (Beltrán Jiménez et al., 2020). Using the covariant formulation of $f(Q)$ gravity can help addressing this challenge. Because of the purely inertial nature of the affine connection in Eq. (13) we can employ a covariant formulation by first establishing the affine connection in a gravity-free setting (Zhao, 2022b).

Moreover, the energy-momentum tensor $E_{\mu\nu}$ can be mathematically expressed as,

$$E_{\mu\nu} = 2 \left(F_{\mu k} F_{\nu k} - \frac{1}{4} g_{\mu\nu} F_{kl} F^{kl} \right). \quad (15)$$

In the above expression,

$$F_{\mu\nu} = \mathcal{A}_{\mu,\nu} - \mathcal{A}_{\nu,\mu}. \quad (16)$$

The electromagnetic field tensor takes the form,

$$F_{\mu\nu,k} + F_{k\mu,\nu} + F_{\nu k,\mu} = 0, \quad (17)$$

$$(\sqrt{-g} F^{\mu\nu})_{,\nu} = \frac{1}{2} \sqrt{-g} J^\mu. \quad (18)$$

The four-potential of the electromagnetic field and the four-current density are denoted by $\mathcal{A}_\mu$ and J^μ respectively. For static fluid distribution exhibiting spherical symmetry, the only nonzero component of the four-current density is J^0 , directed along the radial coordinate r . Thus, except for the radial component F_{01} of the electric field, the remaining elements of the electromagnetic field tensor vanish. If the antisymmetric property $F_{01} = -F_{10}$ is fulfilled, it ensures that Equation (18) holds. The electric field equation is consequently derived from Equation (18) as:

$$E(r) = \frac{1}{2r^2} \int_0^r \Phi(x) \sqrt{g_{rr}} x^2 dx = \frac{q(r)}{r^2}. \quad (19)$$

Here, the charge density and the total charge of the system are symbolized by Φ and $q(r)$ respectively.

3. Modified field equation in $f(Q, T)$ gravity

Our current work is based on static metric configuration having spherical symmetry for depicting the interior structure of compact celestial bodies:

$$ds^2 = -e^{C_0(r)} dt^2 + e^{B_0(r)} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (20)$$

which gives the formula of distance as $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$ with $x^\nu = (t, r, \theta, \phi)$, $C_0(r)$ and $B_0(r)$ corresponding to the space-time coordinates, metric potentials along the time and radial directions, respectively. The dense matter is modelled as isotropic perfect fluid in this work, with the energy-momentum tensor components given as $(-\rho, P, P, P)$. Here matter density, pressure are expressed as ρ and P respectively. The matter Lagrangian is chosen as $L_m = P$ following the works of Negreiros et al. (2009); Ivanov (2010). The expression $\Phi_{\mu\nu} = g_{\mu\nu} P - 2T_{\mu\nu}$ can be used to obtain the components of $\Phi_{\mu\nu}$. The non-metricity scalar is thus given as:

$$Q = -\frac{2e^{-B_0(r)}(rC_0'(r) + 1)}{r^2}. \quad (21)$$

The three independent and non-vanishing field equations in $f(Q, T)$ gravity for the metric (20) are established under the functional form $f(Q, T) = mQ + nT$ as,

$$\rho + \frac{n}{2}(3\rho - P) + \frac{q^2}{r^4} = -m \left[e^{-B_0} \left(\frac{B_0'}{r} - \frac{1}{r} \right) + \frac{1}{r^2} \right], \quad (22)$$

$$P + \frac{n}{2}(3P - \rho) - \frac{q^2}{r^4} = -m \left[e^{-B_0} \left(\frac{C_0'}{r} + \frac{1}{r^2} \right) - \frac{1}{r^2} \right], \quad (23)$$

$$P + \frac{n}{2}(3P - \rho) + \frac{q^2}{r^4} = -m \left[e^{-B_0} \left\{ \frac{C_0''}{2} + \left(\frac{C_0'}{4} + \frac{1}{2r} \right) \times (C_0' - B_0') \right\} \right], \quad (24)$$

and corresponding TOV equation can be recast as:

$$-\frac{C_0'}{2}(\rho + P) - \frac{dP}{dr} - \frac{n}{2(1+n)}(P' - \rho') + \frac{2q}{(n+1)r^4} \frac{dq}{dr} = 0, \quad (25)$$

Next, our aim is to solve the above field equations (22)-(24) and find a most general exact solution that perfectly describes well-behaved models. In this regard, we must find the isotropy condition in $f(Q, T)$ -gravity by subtracting of Eq. (23) from Eq. (24) as, Isotropy conditions is obtained as:

$$-m \left[e^{-B_0} \left\{ \frac{C_0''}{2} + \left(\frac{C_0'}{4} + \frac{1}{2r} \right) (C_0' - B_0') - \left(\frac{C_0'}{r} + \frac{1}{r^2} \right) \right\} + \frac{1}{r^2} \right] = \frac{2q^2}{r^4}. \quad (26)$$

In order to write the above isotropy condition (26) in more trackable, we use the transformation $\mathcal{X}_0 = F_0 r^2$, $e^{C_0} = \mathcal{Y}_0^2$ and $e^{-B_0} = \mathcal{W}_0$ in Eq. (26), we get

$$4\mathcal{X}_0^2 \mathcal{W}_0 \frac{d^2 \mathcal{Y}_0}{d\mathcal{X}_0^2} + 2\mathcal{X}_0^2 \frac{d\mathcal{Y}_0}{d\mathcal{X}_0} \frac{d\mathcal{W}_0}{d\mathcal{X}_0} + \left(\mathcal{X}_0 \frac{d\mathcal{W}_0}{d\mathcal{X}_0} - \mathcal{W}_0 + 1 + \frac{2F_0 q^2}{m\mathcal{X}_0} \right) \mathcal{Y}_0 = 0. \quad (27)$$

and the corresponding expressions for pressure (P) and density (ρ) in terms of $\mathcal{X}_0$, $\mathcal{Y}_0$, $\mathcal{W}_0$ and electric field are:

$$\rho = \frac{1}{\mathcal{X}_0 (2n^2 + 3n + 1) \mathcal{Y}_0} \left[m \left(-2\overline{\mathcal{Y}}_0 \mathcal{X}_0 n \mathcal{W}_0 + \overline{\mathcal{W}}_0 \mathcal{X}_0 \times (3n + 2) \mathcal{Y}_0 + (n + 1) \mathcal{Y}_0 (\mathcal{W}_0 - 1) \right) - \mathcal{E}_2 \mathcal{X}_0 (n + 1) \times \mathcal{Y}_0 \right], \quad (28)$$

$$P = \frac{1}{\mathcal{X}_0 (2n^2 + 3n + 1) \mathcal{Y}_0} \left[-2\overline{\mathcal{Y}}_0 \mathcal{X}_0 m (3n + 2) \mathcal{W}_0 + m \mathcal{Y}_0 (\overline{\mathcal{W}}_0 \mathcal{X}_0 n - n \mathcal{W}_0 + n - \mathcal{W}_0 + 1) + \mathcal{E}_2 \mathcal{X}_0 \times (n + 1) \mathcal{Y}_0 \right], \quad (29)$$

where $\mathcal{E}_2 = \frac{q^2}{r^4}$, $\overline{\mathcal{W}}_0 = \frac{d\mathcal{W}_0}{d\mathcal{X}_0}$, and $\overline{\mathcal{Y}}_0 = \frac{d\mathcal{Y}_0}{d\mathcal{X}_0}$. It is noted that the master equation (27) having three unknown namely $\mathcal{Y}_0$, $\mathcal{W}_0$, and q . Therefore, we use physical viable expression for $\mathcal{W}_0$ as,

$$\mathcal{W}_0 = \frac{7(\mathcal{X}_0 + 1)^2 - 8\mathcal{X}_0(\mathcal{X}_0 + 3)}{7(\mathcal{X}_0 + 1)^2}. \quad (30)$$

After plugging Eq. (30) into Eq. (27), we find

$$\overline{\overline{\mathcal{Y}}}_0 - \frac{8m(\mathcal{X}_0 - 3)\mathcal{X}_0}{2m\mathcal{X}_0(\mathcal{X}_0 + 1)(\mathcal{X}_0^2 + 10\mathcal{X}_0 - 7)} \overline{\mathcal{Y}}_0 - \frac{1}{2m\mathcal{X}_0(\mathcal{X}_0 + 1)(\mathcal{X}_0^2 + 10\mathcal{X}_0 - 7)} \left[7(\mathcal{X}_0 + 1)^3 \times \frac{F_0 q^2}{\mathcal{X}_0^2} + 4\mathcal{X}_0(\mathcal{X}_0 + 5)m \right] \mathcal{Y}_0 = 0, \quad (31)$$

where, $\overline{\overline{\mathcal{Y}}}_0 = \frac{d^2 \mathcal{Y}_0}{d\mathcal{X}_0^2}$. It is evident that the aforementioned Eq. (31) requires an appropriate expression for the electric charge q that would allow Eq. (31) to be fully integrated. We note that when $q = 0$, $\mathcal{Y}_0 = (1 + \mathcal{X}_0)^2$ is a particular solution of Eq. (31). Since, we are interested in finding the non-vanishing expression for electric field. Then we must alter this particular solution and hence by adding a charge parameter σ , we suppose $\mathcal{Y}_0 = (1 + \sigma + \mathcal{X}_0)^2$ is a particular solution. By this assumption, we choose a non-zero formula for the electric charge q as follows:

$$\frac{F_0 q^2}{\mathcal{X}_0^2} = -\frac{4\mathcal{X}_0 m \sigma (2\mathcal{X}_0^2 + \mathcal{X}_0(\sigma + 16) + 5\sigma - 2)}{7(\mathcal{X}_0 + 1)^3 (\mathcal{X}_0 + \sigma + 1)^2}. \quad (32)$$

After plugging the above expression (32) in Eq. (31), we obtain the final form of the master equation as,

$$\overline{\overline{\mathcal{Y}}}_0 (\mathcal{X}_0(\mathcal{X}_0 + 10) - 7)(\mathcal{X}_0 + 1)(\mathcal{X}_0 + \sigma + 1)^2 - 4\overline{\mathcal{Y}}_0 \times (\mathcal{X}_0 - 3)(\mathcal{X}_0 + \sigma + 1)^2 + 8(\mathcal{X}_0 - 3)\sigma \mathcal{Y}_0 - 2(\mathcal{X}_0 + 5) \times (\mathcal{X}_0 + 1)^2 \mathcal{Y}_0 = 0. \quad (33)$$

Now if we look at Eq. (33), the solution $\mathcal{Y}_0 = (1 + \sigma + \mathcal{X}_0)^2$ satisfies the above equation. Then this value of $\mathcal{Y}_0$ is referred as a particular solution of Eq. (33). Hence, the most general solution can be obtained by using change of dependent variable method by assuming the transformation $\mathcal{Y}_0 = \hat{\mathcal{Y}}_0 \mathcal{U}$, where $\hat{\mathcal{Y}}_0 = (1 + \sigma + \mathcal{X}_0)^2$, while $\mathcal{U}$ can be found after integration of Eq. (33) as,

$$\mathcal{U} = \frac{1}{3(\sigma + 1)^3} \left[3\mathcal{A}_1 - \frac{\mathcal{A}_2}{(\sigma^2 - 8\sigma - 16)^2} \right] \times \left(\frac{\sqrt{7 - \mathcal{X}_0(\mathcal{X}_0 + 10)}}{[(\sigma - 8)\sigma - 16](\mathcal{X}_0 + \sigma + 1)^3} \left[3(\mathcal{X}_0 - 3)\sigma^4 + (\mathcal{X}_0(\mathcal{X}_0 + 16) + 119)\sigma^3 + 2(\mathcal{X}_0(5\mathcal{X}_0 + 86) - 47)\sigma^2 + 16(5(\mathcal{X}_0 - 8)\mathcal{X}_0 - 53)\sigma - 96(\mathcal{X}_0 + 1)(3\mathcal{X}_0 + 7) + 3\sigma^5 \right] + \frac{12((\sigma - 4)\sigma(\sigma + 8) + 128)}{(16 - (\sigma - 8)\sigma)^{3/2}} \right]$$

$$\times \tanh^{-1} \left(\frac{\mathcal{X}_0(\sigma - 4) + 5\sigma + 12}{\sqrt{7 - \mathcal{X}_0(\mathcal{X}_0 + 10)}\sqrt{16 - (\sigma - 8)\sigma}} \right) \Bigg), \quad (34)$$

where $\mathcal{A}_1$ and $\mathcal{A}_2$ are constant of integrations. Then, the complete solution of Eq. (30) can be recast as,

$$\begin{aligned} e^{C_0/2} = \mathcal{Y}_0 = \hat{\mathcal{Y}}_0 \mathcal{U} = \frac{((1 + \mathcal{X}_0 + \sigma)^2)}{3(\sigma + 1)^3} \left[- \frac{\mathcal{A}_2}{(\sigma^2 - 8\sigma - 16)^2} \right. \\ \times \left(\frac{\sqrt{7 - \mathcal{X}_0(\mathcal{X}_0 + 10)}}{[(\sigma - 8)\sigma - 16](\mathcal{X}_0 + \sigma + 1)^3} \left[3(\mathcal{X}_0 - 3)\sigma^4 \right. \right. \\ \left. \left. + (\mathcal{X}_0(\mathcal{X}_0 + 16) + 119)\sigma^3 + 2(\mathcal{X}_0(5\mathcal{X}_0 + 86) - 47)\sigma^2 \right. \right. \\ \left. \left. + 16(5(\mathcal{X}_0 - 8)\mathcal{X}_0 - 53)\sigma - 96(\mathcal{X}_0 + 1)(3\mathcal{X}_0 + 7) + 3\sigma^5 \right] \right. \\ \left. + \tanh^{-1} \left(\frac{\mathcal{X}_0(\sigma - 4) + 5\sigma + 12}{\sqrt{7 - \mathcal{X}_0(\mathcal{X}_0 + 10)}\sqrt{16 - (\sigma - 8)\sigma}} \right) \right. \\ \left. \times \frac{12((\sigma - 4)\sigma(\sigma + 8) + 128)}{(16 - (\sigma - 8)\sigma)^{3/2}} \right) + 3\mathcal{A}_1 \Bigg]. \quad (35) \end{aligned}$$

Now we find the expression for density and pressure by substituting values of $\mathcal{Y}_0$ and $\mathcal{W}_0$ in Eqs. (28) and (29) as

$$\begin{aligned} \rho = - \frac{2mF_0}{7(\mathcal{X}_0 + 1)^3 (2n^2 + 3n + 1) \mathcal{Y}_0(\mathcal{X}_0 + \sigma + 1)^2} \\ \times \left[(\mathcal{X}_0 + \sigma + 1)^2 \left(4\mathcal{Y}_0 [\mathcal{X}_0^2(n + 1) + \mathcal{X}_0(n + 2) \right. \right. \\ \left. \left. + 12n + 9] - \mathcal{G}(\mathcal{X}_0) (\mathcal{X}_0^3 + 11\mathcal{X}_0^2 + 3\mathcal{X}_0 - 7)n \right) \right. \\ \left. - 2\mathcal{X}_0(n + 1)\sigma\mathcal{Y}_0 (2\mathcal{X}_0^2 + \mathcal{X}_0(\sigma + 16) + 5\sigma - 2) \right], \quad (36) \end{aligned}$$

$$\begin{aligned} P = \frac{2F_0}{7(\mathcal{X}_0 + 1)^3 (2n^2 + 3n + 1) \mathcal{Y}_0} \left[\mathcal{G}(\mathcal{X}_0)(\mathcal{X}_0 + 1) \right. \\ \times (\mathcal{X}_0^2 + 10\mathcal{X}_0 - 7) m(3n + 2) + 4m\mathcal{Y}_0 [\mathcal{X}_0^2(n + 1) \\ \left. + \mathcal{X}_0(5n + 4) + 3] - \frac{m}{(\mathcal{X}_0 + \sigma + 1)^2} \left(2\mathcal{X}_0(n + 1)\sigma\mathcal{Y}_0 \right. \right. \\ \left. \left. \times (2\mathcal{X}_0^2 + \mathcal{X}_0(\sigma + 16) + 5\sigma - 2) \right) \right], \quad (37) \end{aligned}$$

where, $\mathcal{Y}_0$ is given by Eq. (35), while $\mathcal{G}(\mathcal{X}_0)$ is given in Appendix.

4. Matching condition for the astrophysical system

For any stable spherically symmetric object, the continuity of stellar distribution between the exterior solutions ($r > R$) and the interior ($r < R$) at the boundary ($r = R$) must hold. Such matching conditions ensure a physically viable space-time geometry. Here, the Reissner-Nordstrom exterior metric is adopted to characterize the space-time outside the star:

$$\begin{aligned} dS_+^2 = - \left(1 - \frac{2\mathcal{M}}{r} + \frac{\mathcal{Q}^2}{r^2} + \frac{\Lambda r^2}{3} \right) dt^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \\ + \left(1 - \frac{2\mathcal{M}}{r} + \frac{\mathcal{Q}^2}{r^2} + \frac{\Lambda r^2}{3} \right)^{-1} dr^2, \quad (38) \end{aligned}$$

where the total charge on the boundary $r = R$ is represented by $\mathcal{Q}$ and Λ denotes the cosmological constant. The exterior gravitational field for a charged, non-rotating spherically symmetric mass $\mathcal{M}$ is described by Reissner-Nordstrom metric, which is a solution to the Einstein-Maxwell field equations in static scenario. The most generalized interior metric, incorporating geometric deformation, is given by the following line element as:

$$dS_-^2 = -e^{C_0(r)} dt^2 + e^{B_0(r)} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (39)$$

For a stable system, the inner dS_-^2 (39) and the outer dS_+^2 (38) manifold must be matched smoothly at the boundary Σ . This involves satisfying the well-established continuity equation, ensuring the first and second

fundamental forms align at Σ . Regarding the first fundamental form, the inner metric tensor $g_{\mu\nu}$, defined on the interface by dS_-^2 and dS_+^2 , satisfies:

$$g_{tt}^-|_{r=R} = g_{tt}^+|_{r=R}, \quad (40)$$

$$g_{rr}^-|_{r=R} = g_{rr}^+|_{r=R}. \quad (41)$$

Using Eq. (39) and Eq. (38), the explicit form is derived as,

$$e^{-B_0(R)} = \left(1 - \frac{2\mathcal{M}}{R} + \frac{\mathcal{Q}^2}{R^2} + \frac{\Lambda R^2}{3} \right), \quad (42)$$

$$e^{C_0(R)} = \left(1 - \frac{2\mathcal{M}}{R} + \frac{\mathcal{Q}^2}{R^2} + \frac{\Lambda R^2}{3} \right). \quad (43)$$

Subsequently, the second fundamental form is expressed as,

$$P(R) = 0. \quad (44)$$

It is important to mention that the observed value of the cosmological constant Λ is $10^{-46}/\text{km}^2$, which has a minor influence on the characteristics of compact stars. Consequently, we have excluded the cosmological constant Λ from our analysis of compact stars.

Using the above following conditions, we find the constant parameters as,

$$\begin{aligned} \mathcal{A}_2 = 3(\sigma + 1)^3 \sqrt{\frac{7(\hat{\mathcal{X}}_0 + 1)^2 - 8\hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 3)}{7(\hat{\mathcal{X}}_0 + 1)^2}} \left[3\mathcal{A}_3 \right. \\ \left. + \frac{1}{(\sigma^2 - 8\sigma - 16)^2} \left(\frac{\sqrt{7 - \hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 10)}}{[(\sigma - 8)\sigma - 16](\hat{\mathcal{X}}_0 + \sigma + 1)^3} \right. \right. \\ \left. \times \left[3(\hat{\mathcal{X}}_0 - 3)\sigma^4 + (\hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 16) + 119)\sigma^3 + 2(\hat{\mathcal{X}}_0(5\hat{\mathcal{X}}_0 + 86) - 47)\sigma^2 \right. \right. \\ \left. \left. + 16(5(\hat{\mathcal{X}}_0 - 8)\hat{\mathcal{X}}_0 - 53)\sigma - 96(\hat{\mathcal{X}}_0 + 1)(3\hat{\mathcal{X}}_0 + 7) + 3\sigma^5 \right] \right. \\ \left. + \frac{12((\sigma - 4)\sigma(\sigma + 8) + 128)}{(16 - (\sigma - 8)\sigma)^{3/2}} \right. \\ \left. \times \tanh^{-1} \left(\frac{\hat{\mathcal{X}}_0(\sigma - 4) + 5\sigma + 12}{\sqrt{7 - \hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 10)}\sqrt{16 - (\sigma - 8)\sigma}} \right) \right]^{-1}, \quad (45) \end{aligned}$$

$$\begin{aligned} \mathcal{A}_1 = 3\mathcal{A}_3(\sigma + 1)^3 \sqrt{\frac{7(\hat{\mathcal{X}}_0 + 1)^2 - 8\hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 3)}{7(\hat{\mathcal{X}}_0 + 1)^2}} \left[3\mathcal{A}_3 \right. \\ \left. + \frac{1}{(\sigma^2 - 8\sigma - 16)^2} \left(\frac{\sqrt{7 - \hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 10)}}{[(\sigma - 8)\sigma - 16](\hat{\mathcal{X}}_0 + \sigma + 1)^3} \right. \right. \\ \left. \times \left[3(\hat{\mathcal{X}}_0 - 3)\sigma^4 + (\hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 16) + 119)\sigma^3 + 2(\hat{\mathcal{X}}_0(5\hat{\mathcal{X}}_0 + 86) - 47)\sigma^2 \right. \right. \\ \left. \left. + 16(5(\hat{\mathcal{X}}_0 - 8)\hat{\mathcal{X}}_0 - 53)\sigma - 96(\hat{\mathcal{X}}_0 + 1)(3\hat{\mathcal{X}}_0 + 7) + 3\sigma^5 \right] \right. \\ \left. + \frac{12((\sigma - 4)\sigma(\sigma + 8) + 128)}{(16 - (\sigma - 8)\sigma)^{3/2}} \right. \\ \left. \times \tanh^{-1} \left(\frac{\hat{\mathcal{X}}_0(\sigma - 4) + 5\sigma + 12}{\sqrt{7 - \hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 10)}\sqrt{16 - (\sigma - 8)\sigma}} \right) \right]^{-1}, \quad (46) \end{aligned}$$

where, the symbol $\hat{\mathcal{X}}_0 = F_0 R^2$ and expression for $\mathcal{A}_3$ is given in Appendix.

5. Physical analysis

5.1. Electric charge distribution

For a charged stellar system to be physically viable, it must adhere to the conditions ensuring its relevance, as stated in Delgaty and Lake

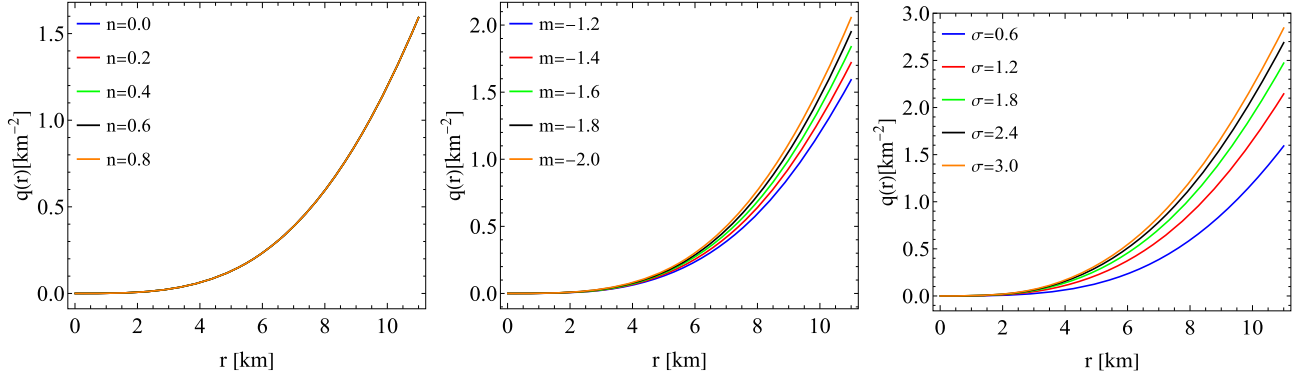


Fig. 1. Influence of $f(Q,T)$ parameters m and n , and charge parameter σ on the electric charge ($q(r)$) versus radial distance r for $F_0 = 0.0021 \text{ km}^{-2}$ with fixed values: (i) $m = -1.2$ and $\sigma = 0.6$ for left panel, (ii) $n = 0.4$ and $\sigma = 0.6$ for middle panel, and (iii) $m = -1.2$ and $n = 0.4$ for right panel.

Table 1

The study of physical values for different $f(Q,T)$ parameter n with fixed values of $F_0 = 0.0021 \text{ km}^{-2}$, $\sigma = 0.6$, and $m = -1.2$.

n	central pressure (P_r) _c ^{eff}	central density ρ_c^{eff}	surface density ρ_s^{eff}	adiabatic index Γ_c	charge (in Coulomb) $Q(C)$
0.0	4.12121×10^{35}	1.3919×10^{15}	7.41301×10^{14}	3.2197	$1.85625 \times 10^{20}C$
0.2	2.64749×10^{35}	1.09357×10^{15}	5.70232×10^{14}	2.92166	$1.85625 \times 10^{20}C$
0.4	1.96906×10^{35}	8.97578×10^{14}	4.63313×10^{14}	2.81108	$1.85625 \times 10^{20}C$
0.6	1.57993×10^{35}	7.60582×10^{14}	3.90159×10^{14}	2.75845	$1.85625 \times 10^{20}C$
0.8	1.32617×10^{35}	6.59739×10^{14}	3.36955×10^{14}	2.72932	$1.85625 \times 10^{20}C$

Table 2

The study of physical values for different $f(Q,T)$ parameter m with fixed values of $F_0 = 0.0021 \text{ km}^{-2}$, $\sigma = 0.6$, and $n = 0.4$.

m	central pressure (P_r) _c ^{eff}	central density ρ_c^{eff}	surface density ρ_s^{eff}	adiabatic index Γ_c	charge (in Coulomb) $Q(C)$
-1.2	1.96906×10^{35}	8.97282×10^{14}	4.63313×10^{14}	2.81108	$1.85625 \times 10^{20}C$
-1.4	2.29723×10^{35}	1.04683×10^{15}	5.40532×10^{14}	2.81108	$2.00498 \times 10^{20}C$
-1.6	2.62541×10^{35}	1.19638×10^{15}	6.17751×10^{14}	2.81108	$2.14342 \times 10^{20}C$
-1.8	2.95358×10^{35}	1.34592×10^{15}	6.9497×10^{14}	2.81108	$2.27344 \times 10^{20}C$
-2.0	3.28176×10^{35}	1.49547×10^{15}	7.72189×10^{14}	2.81108	$2.39641 \times 10^{20}C$

Table 3

The effect of charge parameter σ on physical parameters with fixed values of $F_0 = 0.0021 \text{ km}^{-2}$, $m = -1.2$, and $n = 0.4$.

σ	central pressure (P_r) _c ^{eff}	central density ρ_c^{eff}	surface density ρ_s^{eff}	adiabatic index Γ_c	charge (in Coulomb) $Q(C)$
0.6	1.96906×10^{35}	8.97282×10^{14}	4.63313×10^{14}	2.81108	$1.85625 \times 10^{20}C$
1.2	1.67086×10^{35}	8.93141×10^{14}	4.5858×10^{14}	2.66431	$2.50052 \times 10^{20}C$
1.8	1.4502×10^{35}	8.90077×10^{14}	4.55124×10^{14}	2.53522	$2.88129 \times 10^{20}C$
2.4	1.28713×10^{35}	8.87813×10^{14}	4.52556×10^{14}	2.4305	$3.13441 \times 10^{20}C$
3.0	1.16329×10^{35}	8.86093×10^{14}	4.5059×10^{14}	2.34476	$3.31514 \times 10^{20}C$

(1998). The behavior of charge ($q(r)$) was tested by variation with the parameters n , m , and σ . In Fig. 1, it was found that the charge is finite and regular at the center, which fulfills a fundamental condition for the physical acceptance of the stellar model (Varela et al., 2010). It was observed that, the charge vanishes at the center and keeps on increasing in a radially outward direction, attaining its maxima at the stellar boundary. The surface charge was observed to be affected by the charge parameter σ and m while being unaffected by n . Specifically, as the charge increases with an increase in σ and m , while it remains the same for n . The aforementioned findings suggest that the charge and nonmetricity parameters significantly influence the electric field. The values of surface charge are indicated in Tables 1, 2, and 3. The values of charge are ranging between $[1.85, 3.31] \times 10^{20}C$ which fit well with the prior studies. The studies done in Ref. Varela et al. (2010) suggest that the surface charge of the order $\approx 10^{19}C$ can signify that the

charge is dissipated by the stars along their surface, while for the surface charge of the order $\approx 10^{20}C$, it can be sustained by neutron stars to prevent gravitational collapse (Ray et al., 2003), by counterbalancing the attractive gravitational force by charge repulsion (Sharma et al., 2001; de Felice et al., 1995).

5.2. Physical behavior of density and pressure

The density, $\rho(r)$, and isotropic pressure, $P(r)$, of the stellar configuration were subsequently studied. From Fig. 2 and 3, it is evident that the central values of density and radial pressure are both finite, ensuring the regularity and non-singularity at the core. Furthermore, both density and pressure exhibit a decreasing trend as they move outward, attaining maximum values at the center and minimum values at the surface. The variation of $\rho(r)$ and $P(r)$ with respect to σ , m , and n

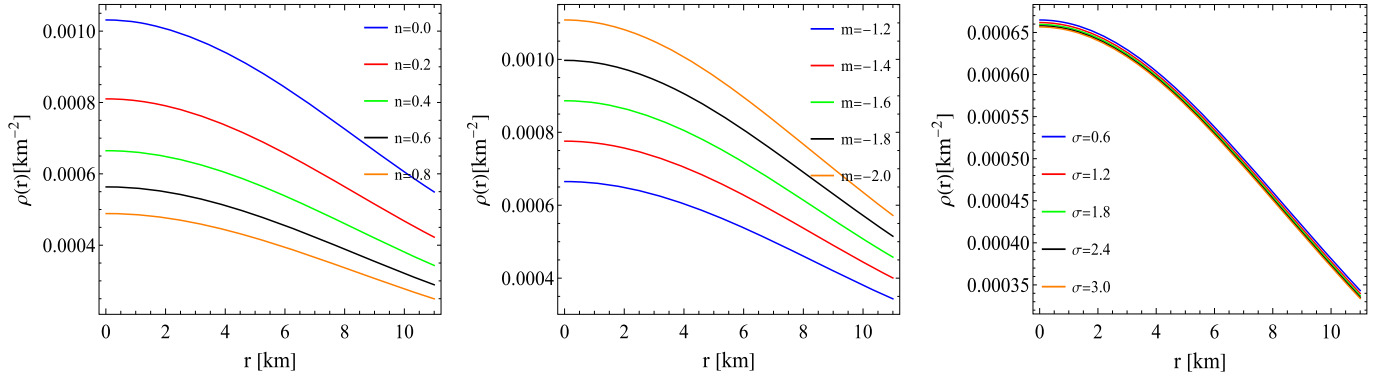


Fig. 2. Influence of $f(Q,T)$ parameters m and n , and charge parameter σ on the energy density ($\rho(r)$) versus radial distance r for $F_0 = 0.0021 \text{ km}^{-2}$ with fixed values: (i) $m = -1.2$ and $\sigma = 0.6$ for left panel, (ii) $n = 0.4$ and $\sigma = 0.6$ for middle panel, and (iii) $m = -1.2$ and $n = 0.4$ for right panel.

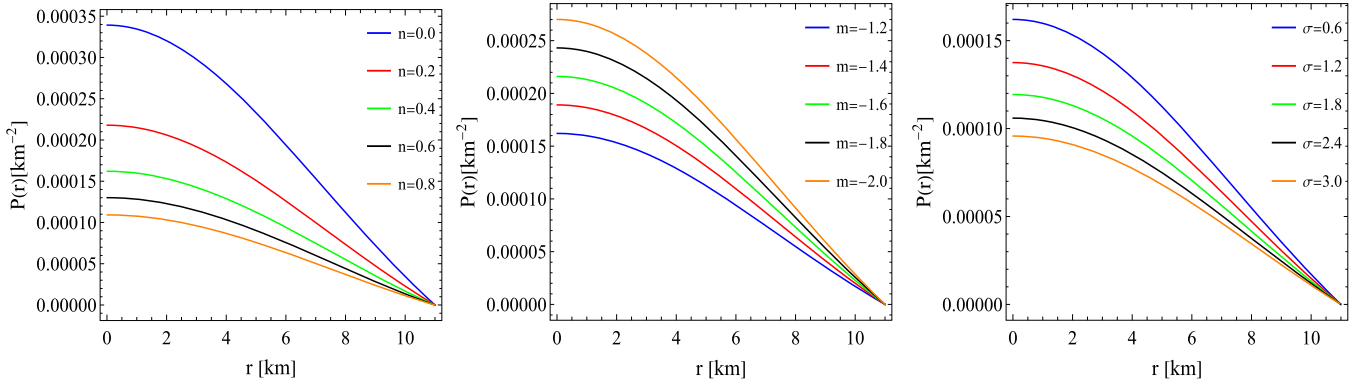


Fig. 3. Influence of $f(Q,T)$ parameters m and n , and charge parameter σ on the pressure ($P(r)$) versus radial distance r for $F_0 = 0.0021 \text{ km}^{-2}$ with fixed values: (i) $m = -1.2$ and $\sigma = 0.6$ for left panel, (ii) $n = 0.4$ and $\sigma = 0.6$ for middle panel, and (iii) $m = -1.2$ and $n = 0.4$ for right panel.

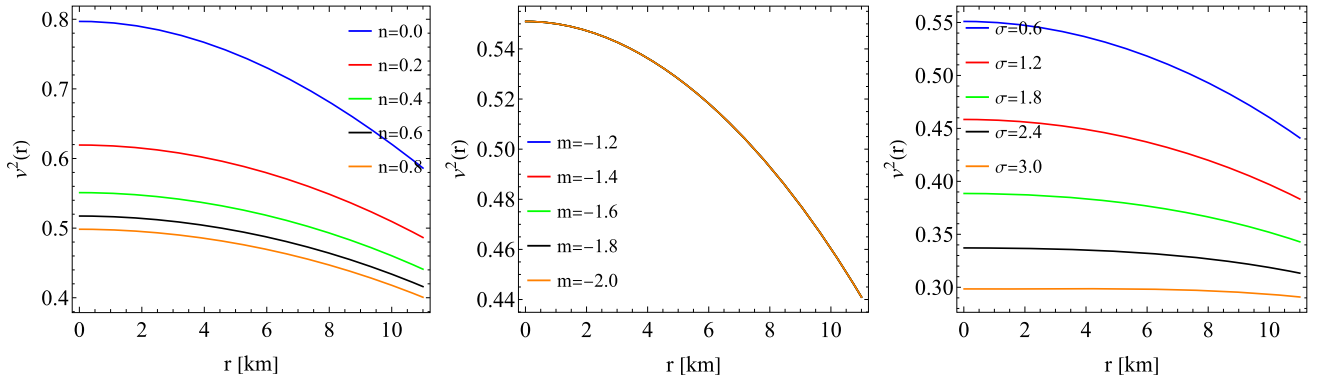


Fig. 4. Influence of $f(Q,T)$ parameters m and n , and charge parameter σ on the sound velocity ($v^2(r)$) versus radial distance r for $F_0 = 0.0021 \text{ km}^{-2}$ with fixed values: (i) $m = -1.2$ and $\sigma = 0.6$ for left panel, (ii) $n = 0.4$ and $\sigma = 0.6$ for middle panel, and (iii) $m = -1.2$ and $n = 0.4$ for right panel.

shows that $\rho(r)$ remains largely unaffected by σ , while $P(r)$ decreases with an increase in σ . However, both $\rho(r)$ and $P(r)$ decrease with increasing n , but both increase with an increase with a rise in m . The pressure becomes zero at the center for all values of σ , m , and n , fulfilling the vanishing condition of isotropic pressure. From Tables 1, 2, and 3, it is noted that the density lies within the order of 10^{14} g/cm^3 to 10^{15} g/cm^3 , and the subsequent pressure has a magnitude of the order $10^{35} \text{ dyne/cm}^2$. The central pressure is observed to be ranging in between $[1.16, 4.12] \times 10^{35} \text{ dyne/cm}^2$, while the central density was found to be in the range of $[6.59 \times 10^{14}, 1.49 \times 10^{15}] \text{ g/cm}^3$, as seen in Tables 1, 2, and 3. Such densities are indicative of neutron-rich composition, possibly including electrons and muons (Ruderman, 1972). This aligns well with the isotropic stellar systems in the GR framework, as

shown by some earlier works (Maurya et al., 2015; Dayanandan et al., 2020).

5.3. Causality condition

For any physically acceptable stellar model, the fulfillment of the causality condition is an essential criterion. This means the sound speed within the star must remain below the speed of light, i.e. $v^2(r) < 1$ (Herrera, 1992), where $c = 1$. The sound speed is expressed as

$$v^2(r) = \frac{dP}{d\rho}. \quad (47)$$

Fig. 4 demonstrates variations in sound speed across the stellar structure for different values of n , m , and σ . The general nature of the

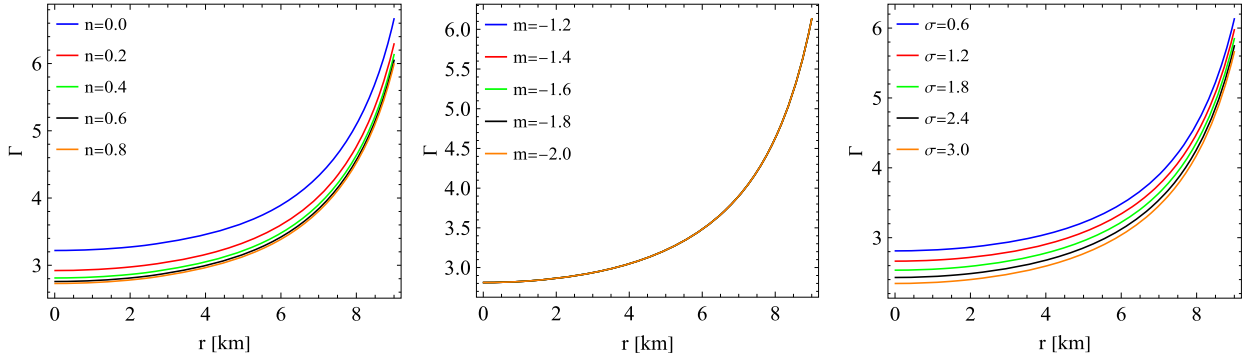


Fig. 5. Influence of $f(Q,T)$ parameters m and n , and charge parameter σ on the adiabatic index ($\Gamma(r)$) versus radial distance r for $F_0 = 0.0021 \text{ km}^{-2}$ with fixed values: (i) $m = -1.2$ and $\sigma = 0.6$ for left panel, (ii) $n = 0.4$ and $\sigma = 0.6$ for middle panel, and (iii) $m = -1.2$ and $n = 0.4$ for right panel.

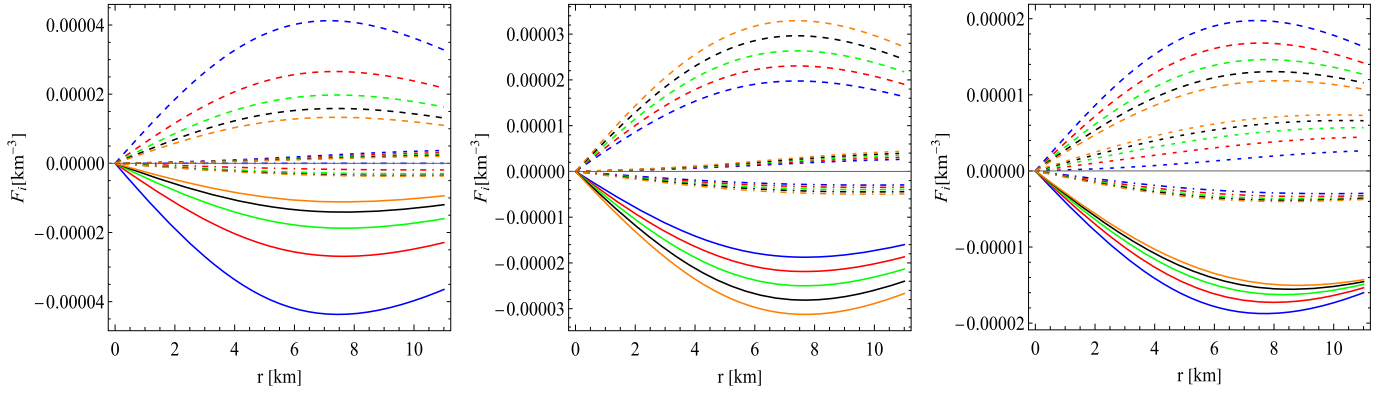


Fig. 6. The hydrostatic equilibrium of $f(Q,T)$ models. The figures show the variation of different forces as: F_g -solid lines, F_h -dashed lines, $F_{(Q,T)}$ -Dot-dashed lines, F_q -small-dashed lines. The color description as follows: (i) Left Panel: $n = 0.0$ -blue, $n = 0.2$ -red, $n = 0.4$ -green, $n = 0.6$ -black, $n = 0.8$ -orange with fixed $\sigma = 0.6$, and $m = -1.2$ (ii) Middle Panel: $m = -1.2$ -blue, $m = -1.4$ -red, $m = -1.6$ -green, $m = -1.8$ -black, $m = -2.0$ -orange with fixed $\sigma = 0.6$, and $n = 0.4$, (iii) Right Panel: $\sigma = 0.6$ -blue, $\sigma = 1.2$ -red, $\sigma = 1.8$ -green, $\sigma = 2.4$ -black, $\sigma = 3.0$ -orange with fixed $m = -1.2$, and $n = 0.4$.

curves indicates that $v^2(r)$ attains its maxima at the center and decreases radially throughout. We observe that all the cases fulfill the above-mentioned condition, exhibiting distinct trends for each parameter. For fixed m and σ , $v^2(r)$ seems to be decreasing with an increment of n , remains unchanged with the parameter m , and it elevates with decrements in σ . As the parameters n and σ significantly impact the sound speed, which means these parameters impact the stiffness of the equation of state (EOS) (Canuto, 1973). The EOS is stiffer when the sound speed is faster. This means that energy-momentum coupling and charge both make the EOS softer, but the nonmetricity parameter m does not change it.

5.4. Adiabatic index: stability analysis

The stability in terms of the adiabatic index $\Gamma(r)$ must be checked to assess stability of the system. For isotropic matter distribution, the adiabatic index is given as,

$$\Gamma = \left(\frac{P + \rho}{P} \right) \left[\frac{dP}{d\rho} \right]. \quad (48)$$

The stability criterion in this regard is given as $\Gamma > 4/3$ (Chandrasekhar, 1964; Wald, 1984). The variation of Γ is shown in Fig. 5, and it was found that this condition is met in all cases. The general trend of the variation of $\Gamma(r)$ shows that it initially increases very slightly near the core in a radially outward direction, but near the surface it increases in an accelerated manner. Moreover, it is also seen that Γ is not influenced by the parameter m , while it is significantly influenced by both parameters n and σ , implying that the charge and energy-momentum coupling parameters play a crucial role in the stability of the system in terms of the adiabatic index. As the adiabatic index Γ decreases with an

increase in σ , it can be said that at a certain value of σ , the adiabatic index may fall below the stability limit. Which in turn signifies, that there is a limiting value of maximum charge to maintain stability, and the charge cannot be arbitrarily large. In our investigation, when we fixed the values of $f(Q,T)$ parameters m and n , then we found that the system will remain stable for higher values of the charge parameter σ and amount of total charge as well as mass-radius ratio will not go beyond the order 10^{20} C and $4/9$, respectively. For fixing $m = -1.2$ and $n = 0.4$, the adiabatic index $\Gamma \geq 4/3$ and $Q \approx 4.32916 \times 10^{20} \text{ C}$ for $\sigma \leq 62$.

5.5. Hydrostatic balance

To examine the hydrostatic equilibrium of the $f(Q,T)$ model, it is essential to investigate the modified Tolman-Oppenheimer-Volkoff (TOV) equation,

$$-\frac{C'_0}{r}(\rho + P) - \frac{dP}{dr} - \frac{n}{2(1+n)}(P' - \rho') + \frac{2q}{(n+1)r^4} \frac{dq}{dr} = 0, \quad (49)$$

The modified TOV equation (49) can be decomposed into the following different forces,

$$F_g = -\frac{C'_0}{2}(\rho + P), F_h = -\frac{dP}{dr}, F_{(Q,T)} = -\frac{n}{2(1+n)}(P' - \rho'), \quad (50)$$

$$F_q = \frac{2q}{(n+1)r^4} \frac{dq}{dr}.$$

Fig. 6 shows the hydrostatic equilibrium in terms of different forces, such as the electrostatic force F_q , the gravitational force F_g , the hydrostatic force F_h , and the coupling force $F_{(Q,T)}$. For a stable system, the forces must cancel each other, i.e., $F_g + F_h + F_{(Q,T)} + F_q = 0$. It is observed that the negative and attractive F_g and $F_{(Q,T)}$ are balanced by

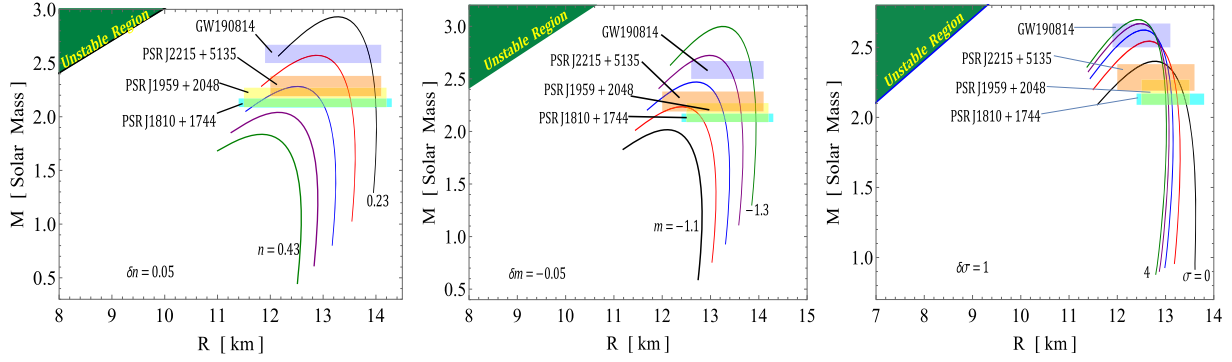


Fig. 7. Influence of $f(Q,T)$ parameters m and n , and charge parameter σ on mass-radius relation for the same values as used in Fig. 6.

Table 4

The predicted radii of few high mass compact stars for different values of m and n .

Objects	$\frac{M}{M_{\odot}}$	Predicted R (km)				Predicted R (km)		
		m				n		
		-1.15	-1.20	-1.25	-1.30	0.23	0.28	0.33
PSR J1810+1744 (Romani et al., 2021b)	2.13 ± 0.04	$12.81^{+0.05}_{-0.06}$	$13.29^{+0.02}_{-0.05}$	$13.64^{+0.01}_{-0.01}$	$13.94^{+0.01}_{-0.01}$	$14.00^{+0.01}_{-0.02}$	$13.53^{+0.01}_{-0.04}$	$12.98^{+0.05}_{-0.12}$
PSR J1959+2048 (Kandel and Romani, 2020)	2.18 ± 0.09	$12.70^{+0.16}_{-0.06}$	$13.26^{+0.05}_{-0.06}$	$13.63^{+0.02}_{-0.03}$	$13.93^{+0.01}_{-0.01}$	$13.99^{+0.01}_{-0.01}$	$13.50^{+0.04}_{-0.04}$	$12.91^{+0.12}_{-0.25}$
PSR J2215+5135 (Kandel and Romani, 2020)	$2.28^{+0.10}_{-0.09}$	—	$13.19^{+0.07}_{-0.12}$	$13.61^{+0.02}_{-0.05}$	$13.92^{+0.01}_{-0.01}$	$13.97^{+0.02}_{-0.03}$	$13.45^{+0.04}_{-0.07}$	—
GW190814 (Abbott et al., 2020d)	$2.5-2.67$	—	—	$13.42^{+0.06}_{-0.14}$	$13.86^{+0.02}_{-0.05}$	$13.86^{+0.03}_{-0.05}$	$12.92^{+0.28}_{-0}$	—

positive and repulsive F_h and F_q . The force F_g is dominant over $F_{(Q,T)}$ and F_h is dominant over F_q . The variation of forces shows clear patterns. The variation of forces exhibits distinct trends. The gravitational force F_g decreases with n and σ , but increases with m . The coupling force $F_{(Q,T)}$ vanishes for $n=0$ and becomes increasingly negative as n grows, with its magnitude also increasing. Additionally, $F_{(Q,T)}$ strengthens with higher values of m and σ . The electrostatic force F_q decreases as n increases but grows with increasing m and σ . Similarly, the hydrostatic force F_h decreases with n and σ , while increasing with m . From the above analysis, it can be interpreted that m , or the matter part in the non-minimal coupling enhances the coupling force while minimizing all the other forces. In contrast, the parameter n associated to nonmetricity enhances all the forces. However, the charge parameter σ reduces the higher magnitude forces such as the gravitational force F_g and hydrostatic force F_h , while enhancing the lower magnitude forces such as the coupling force and $F_{(Q,T)}$, electrostatic force F_q . This effectively means that the charge parameter σ plays a significant role in balancing the forces, enhancing the stability of the system.

5.6. Mass-radius relationship of compact stars in $f(Q,T)$ gravity

Using observational data of mass-radius of compact stars, the authors in Maurya et al. (2023) indicated that it is possible to generate compact star masses beyond $2M_{\odot}$ using $f(Q)$ gravity. In another work, the authors (Maurya et al., 2024) showed that $f(Q)$ gravity supports the existence of a broad spectrum of the observed range of masses in $2.08 - 2.67M_{\odot}$ region, where the upper limit falls in the mass gap region. The incorporation of a nonmetricity tensor alongside the trace of the energy-momentum tensor in $f(Q,T)$ gravity makes it a viable framework for describing ultra-dense compact objects. Moreover, the mass-radius alongside many other properties and stability of the stellar structure can be comprehensively described by the modified field equations with effects of non-metricity. The viability of $f(Q,T)$ gravity can be tested by its efficiency in predicting the mass-radius relation, by comparing with the observational counterpart.

The mass-radius relationship of the charged isotropic compact star model is tested by examining the influence of the coupling parameters m and n alongside the charge parameter σ . In Fig. 7, it is observed that the mass (M) initially increases radially, then reaches a maximum point,

and starts to decrease. The peak value of mass corresponds to the maximum allowable mass in $f(Q,T)$ gravity. It is also observed that the maximum value of mass decreases with n but increases with the increase in magnitude of m and σ . Which essentially means, the maximum allowable mass can attain higher values for charged distribution and the effect of non-metricity. So the combined and contrasting effect of m and n ensures stability while helping the stellar system to attain sustainable maximum mass. It is also noticed that the current stellar model satisfies the Buchdahl limit for all combinations of the parameters.

Predicting the radii of compact stars are often trickier than obtaining mass and is often done by comparison with observational data (Fortin et al., 2015; Özel and Freire, 2016). In this case the radii are calculated by using the observed data from the pulsars such as PSR J1810+1744 (Romani et al., 2021a), PSR J1959+2048 (Kandel et al., 2021), PSR J2215+5135 (Linares et al., 2018), alongside a secondary companion in GW190814 (Abbott et al., 2020c) as the compact objects with known mass. The predicted radii for the observed stars are listed in Tables 4 and 5 for different values of n , m and σ , which shows that the predicted radii range in between ([12.70, 14.00]) km, which fits well with the obtained radius of several neutron star EOS in Lattimer and Prakash (2001).

The maximum values of M for various parameters are listed in Table 6, which shows that the maximum value of mass stays within the range of ($[1.83, 2.99]M_{\odot}$), with the subsequent radii ranging between ($[11.91, 13.30])$ km. Several observed stars such as GW190814, PSR J0740+6620 falls within this range (Fonseca et al., 2021; Abbott et al., 2020c). In Nashed and Harko (2024), the authors argued that the maximum compactness is 10% higher in $f(Q,T)$ gravity compared to its GR counterpart, because of the strong energy condition. In our article, the maximum obtained mass of $2.99M_{\odot}$ with a subsequent radius of 13.29 backs up their study. From Tables 4 and 5 it is noted that the predicted radius of the compact stars decreases as the mass increases. Which means, the compactness increases the value of mass. This implies that higher mass stars can exist in a more compact configuration. It is also noted that the radius decreases with charge parameter σ and the coupling parameter n , while it increases with the parameter associated with nonmetricity m . The parameters m and n influence mass and radius in a similar manner, so, varying either one alone does not significantly enhance compactness. However, as σ effectively increases mass as well as decreases radius, it can be inferred that the existence of electric

Table 5
The predicted radii of few high mass compact stars for different values of σ .

Objects	$\frac{M}{M_\odot}$	Predicted R (km)				
		σ				
		0	1	2	3	4
PSR J1810+1744 (Romani et al., 2021b)	2.13 ± 0.04	$13.40^{+0.03}_{-0.03}$	$13.24^{+0.02}_{-0.01}$	$13.13^{+0.01}_{-0.01}$	$13.05^{+0.01}_{-0.01}$	$13.00^{+0.01}_{-0.01}$
PSR J1959+2048 (Kandel and Romani, 2020)	2.18 ± 0.09	$13.36^{+0.07}_{-0.10}$	$13.23^{+0.03}_{-0.05}$	$13.12^{+0.01}_{-0.03}$	$13.04^{+0.02}_{-0.02}$	$12.99^{+0.02}_{-0.02}$
PSR J2215+5135 (Kandel and Romani, 2020)	$2.28^{+0.10}_{-0.09}$	$13.24^{+0.11}_{-0.24}$	$13.17^{+0.05}_{-0.16}$	$13.08^{+0.05}_{-0.05}$	$13.02^{+0.02}_{-0.04}$	$12.97^{+0.02}_{-0.04}$
GW190814 (Abbott et al., 2020d)	$2.5\text{--}2.67$	—	—	$12.81^{+0.12}_{-}$	$12.82^{+0.08}_{-}$	$12.81^{+0.05}_{-}$

Table 6
Maximum masses and corresponding radii for different values of n , m and σ .

n	$M_{\max}/M_\odot$	R (km)	m	$M_{\max}/M_\odot$	R (km)	σ	$M_{\max}/M_\odot$	R (km)
0.23	2.93	13.30	−1.10	2.02	12.11	0	2.40	12.79
0.28	2.57	12.91	−1.15	2.23	12.46	1	2.54	12.68
0.33	2.28	12.54	−1.20	2.46	12.75	2	2.62	12.55
0.38	2.04	12.20	−1.25	2.72	13.03	3	2.67	12.49
0.43	1.83	11.91	−1.30	2.99	13.29	4	2.70	12.44

charge increases compactness, supporting the existence of higher-mass compact stars falling in the mass gap. In this connection, the radius of the secondary companion in GW190814, one of the compact objects in the mass gap region, is predicted to fall in the range [12.81, 13.89] km by the present isotropic charged stellar model in $f(Q, T)$ gravity.

6. Concluding remarks

In this study, the charged isotropic stellar model within the framework of $f(Q, T)$ gravity was examined. After obtaining the solutions for the modified field equations, the physical properties of the system were analyzed to verify physical relevance and stability. The analysis revealed that the fundamental criteria of regularity and finiteness are satisfied by the electric charge distribution, as shown in Fig. 1. The behavior of charge was observed to be significantly influenced by the charge parameter σ and the coupling parameter m , while it remained unaffected by n . The surface charge was found to be within the range of $1.85 \times 10^{20} C$ to $3.31 \times 10^{20} C$, fitting well with earlier findings, as seen in Tables 1, 2, and 3. These findings indicate that the inclusion of charge plays a crucial role in the stability of the stellar structure by counteracting gravitational collapse.

Following this, the density and isotropic pressure were studied. It was found that both are finite and regular at the center, making sure that there are no singularities (Figs. 2 and 3). Tables 1, 2, and 3 revealed that the central density ranges in between 6.59×10^{14} and 1.49×10^{15} in this study, which is consistent with neutron star configurations. The isotropic pressure was observed to vanish at the boundary, which is an essential criterion of physical relevance. Variations in the parameters n , m , and σ revealed that while $\rho(r)$ is relatively unaffected by σ , $P(r)$ decreases with its increase.

Next, the causality condition was tested and it was found from Fig. 4 that the sound speed remains subluminal, fulfilling the required criteria for physical relevance. Similarly, another key indicator of stability, the adiabatic index $\Gamma(r)$, was shown to exceed the critical value of $4/3$ throughout the configuration (Fig. 5). Moreover, both the sound speed and adiabatic index were shown to be sensitive to variations in n and σ , but not with m . Which implies that the charge and energy-momentum coupling play a significant role in these parameters, while they aren't affected by nonmetricity.

The equilibrium of forces within the system was analyzed using modified TOV equations in terms of four different forces, as shown in Fig. 6. It was observed that the attractive gravitational force and the coupling force $F_{(Q,T)}$ are balanced by the repulsive electrostatic and hydrostatic forces.

The mass-radius relationship was studied to evaluate the model's viability in predicting the properties of compact stars. The maximum mass, was found to range between $1.83 M_\odot$ and $2.99 M_\odot$, with corresponding radii between 11.91 km and 13.30 km. Also, the maximum value of mass was found to be decreasing with increasing n and increasing with m and σ . These results satisfy the Buchdahl limit and are consistent with observational data. The predicted radii for the observed stars listed in Tables 4 and 5 for different values of n , m and σ shows that the estimated radii range in between ([12.70, 14.00]) km, which align well with the observations. As demonstrated by the authors (Nashed and Harko, 2024), that due to the strong energy condition, the maximum compactness in $f(Q, T)$ gravity exceeds that of General Relativity (GR) by 10%. Our findings support this claim, as we obtain a maximum mass of $2.99 M_\odot$ with a corresponding radius of 13.29, consistent with their results.

The role of charge parameter σ indicated that it enhances compactness by increasing mass and decreasing radius. And as it is already observed that higher-mass compact stars have lower radii, which essentially means increasing compactness can support higher-mass compact stars falling in the mass gap region. So, by increasing compactness, the presence of electric charge can support the existence of higher-mass compact stars. Moreover, σ seemingly influences the hydrostatic balance of the system by enhancing the lower magnitude forces such as the electrostatic force F_q and coupling force $F_{(Q,T)}$, and reducing the higher magnitude forces such as hydrostatic force F_h and gravitational force F_g , enhancing the stability of the system. However, although the presence of charge can support higher mass compact stars as well as enhances hydrostatic balance, it can be arbitrarily large, as the adiabatic index $\Gamma(r)$ decreases with σ , indicating there is a higher limit of charge, beyond which the adiabatic index goes below the stable limit of $4/3$.

In conclusion, it is noteworthy that an isotropic charged stellar configuration has been effectively investigated within the framework of $f(Q, T)$ gravity, with a comprehensive investigation of the effects of non-minimal coupling between the geometric component Q and the matter component T . The present method can be used to investigate some more most general exact solutions for compact stars for anisotropic/charge matter distribution in many other modified gravity theories such as $f(R, T)$, $f(Q)$ and $f(R, L_m)$ including $f(Q, T)$ gravity theories etc.

CRediT authorship contribution statement

S.K. Maurya: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.
A. Errehymy: Writing – review & editing, Writing – original draft,

Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Z. Umbetova:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **K. Myrzakulov:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **I. Ibragimov:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **A. Dauletov:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **J. Rayimbaev:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no competing interests that could influence the results or interpretation of this research. They have no

financial, personal, or professional affiliations that might be perceived as a bias in the conduct or reporting of the study. This includes any financial relationships with organizations or entities that have a direct interest in the subject matter discussed in this manuscript. All authors have read and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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Appendix A

$$\begin{aligned}
 G(\mathcal{X}_0) &= \frac{2(\mathcal{X}_0 + \sigma + 1)}{3(\sigma + 1)^3} \left[-\frac{\mathcal{A}_2}{(\sigma^2 - 8\sigma - 16)^2} \left(\frac{\sqrt{7 - \mathcal{X}_0(\mathcal{X}_0 + 10)} \mathcal{G}_1(\mathcal{X}_0)}{((\sigma - 8)\sigma - 16)(\mathcal{X}_0 + \sigma + 1)^3} + \frac{12((\sigma - 4)\sigma(\sigma + 8) + 128)}{(16 - (\sigma - 8)\sigma)^{3/2}} \right. \right. \\
 &\quad \left. \left. \times \tanh^{-1} \left(\frac{\mathcal{X}_0(\sigma - 4) + 5\sigma + 12}{\sqrt{7 - \mathcal{X}_0(\mathcal{X}_0 + 10)} \sqrt{16 - (\sigma - 8)\sigma}} \right) \right) + 3\mathcal{A}_1 \right] + \frac{\mathcal{A}_2(\mathcal{X}_0 + 1)}{\sqrt{-\mathcal{X}_0^2 - 10\mathcal{X}_0 + 7(\sigma + 1)^3(\mathcal{X}_0 + \sigma + 1)^2}}, \\
 G_1(\mathcal{X}_0) &= 3(\mathcal{X}_0 - 3)\sigma^4 + (\mathcal{X}_0(\mathcal{X}_0 + 16) + 119)\sigma^3 + 2(\mathcal{X}_0(5\mathcal{X}_0 + 86) - 47)\sigma^2 + 16(5(\mathcal{X}_0 - 8)\mathcal{X}_0 - 53)\sigma - 96(\mathcal{X}_0 + 1) \\
 &\quad \times (3\mathcal{X}_0 + 7) + 3\sigma^5, \\
 \mathcal{A}_3 &= \frac{4(\sigma^3 + 4\sigma^2 - 32\sigma + 128)}{(-\sigma^2 + 8\sigma + 16)^{7/2}} \tanh^{-1} \left(\frac{\hat{\mathcal{X}}_0(\sigma - 4) + 5\sigma + 12}{\sqrt{7 - \hat{\mathcal{X}}_0(\hat{\mathcal{X}}_0 + 10)} \sqrt{16 - (\sigma - 8)\sigma}} \right) + \sqrt{-\hat{\mathcal{X}}_0^2 - 10\hat{\mathcal{X}}_0 + 7} \left[2\hat{\mathcal{X}}_0^6(5n + 4) \right. \\
 &\quad \times (\sigma^3 + 10\sigma^2 + 80\sigma - 288) + 8\hat{\mathcal{X}}_0^5(5n(\sigma^4 + 9\sigma^3 + 88\sigma^2 - 32\sigma - 960) + 4\sigma^4 + 35\sigma^3 + 342\sigma^2 - 208\sigma - 3552) \\
 &\quad + 2\hat{\mathcal{X}}_0^4(n(31\sigma^5 + 236\sigma^4 + 2809\sigma^3 + 8562\sigma^2 - 47728\sigma - 69792) + 25\sigma^5 + 172\sigma^4 + 2132\sigma^3 + 5568\sigma^2 - 33600\sigma - 52224) \\
 &\quad + \hat{\mathcal{X}}_0^3[n(45\sigma^6 + 282\sigma^5 + 3788\sigma^4 + 29552\sigma^3 - 68960\sigma^2 - 325632\sigma - 225792) + 2(18\sigma^6 + 97\sigma^5 + 1394\sigma^4 + 9952\sigma^3 \\
 &\quad - 20208\sigma^2 - 119296\sigma - 89856)] + \hat{\mathcal{X}}_0^2[n(15\sigma^7 + 69\sigma^6 + 878\sigma^5 + 17268\sigma^4 - 4258\sigma^3 - 254164\sigma^2 - 414624\sigma - 180672) \\
 &\quad + 2(6\sigma^7 + 15\sigma^6 + 419\sigma^5 + 5782\sigma^4 + 3204\sigma^3 - 92600\sigma^2 - 171456\sigma - 81024)] + \hat{\mathcal{X}}_0[n(48\sigma^7 - 309\sigma^6 + 2438\sigma^5 \\
 &\quad + 14172\sigma^4 - 57240\sigma^3 - 212064\sigma^2 - 210688\sigma - 67584) + 2(15\sigma^7 - 36\sigma^6 + 807\sigma^5 + 7702\sigma^4 - 20332\sigma^3 - 97128\sigma^2 \\
 &\quad - 107200\sigma - 37248)] + (\sigma + 1) \left\{ 3n(3\sigma^6 - 114\sigma^5 + 558\sigma^4 - 898\sigma^3 - 5596\sigma^2 - 6560\sigma - 2880) + 2(21\sigma^6 - 150\sigma^5 \right. \\
 &\quad + 1218\sigma^4 - 748\sigma^3 - 11248\sigma^2 - 15680\sigma - 6912) \left. \right\} \left[6(\sigma^2 - 8\sigma - 16)^3(\hat{\mathcal{X}}_0 + \sigma + 1)^3 \left[\hat{\mathcal{X}}_0^4(5n + 4) + \hat{\mathcal{X}}_0^3(5n(\sigma + 10) \right. \right. \\
 &\quad + 4(\sigma + 9)) + \hat{\mathcal{X}}_0^2(n(\sigma^2 + 41\sigma + 64) + \sigma^2 + 26\sigma + 52) + \hat{\mathcal{X}}_0(n(5\sigma^2 + 31\sigma - 2) + 3(\sigma^2 + 12\sigma + 4)) - (\sigma + 1) \right. \\
 &\quad \left. \left. \times (21n - 6\sigma + 8) \right] \right]^{-1}.
 \end{aligned}$$

Data availability

No data was used for the research described in the article.

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